Characters & Strings

Handling Text in Java Programs

char

Primitive type for **single** characters.

Notation: **'Single Quotes'**

```
char grade = 'A';
char sym = '$';
```

String

Non-primitive type for **sequences** (text).

Notation: **"Double Quotes"**

```
String name = "Alice";
String msg = "Hello!";
```

1. The ASCII Connection

Characters are secretly stored as numbers. You can use ASCII values directly:

```
char a = 65; // Result: 'A'
char b = 66; // Result: 'B'
```

2. Escape Sequences

Special characters that cannot be typed normally:

`\n : New Line`	`\t : Tab`	`\" : Double Quote`
`\' : Single Quote`	`\\ : Backslash`	`\r : Return`

3. String Power (Methods)

Since `String` is an object, it comes with built-in "superpowers" (methods):

.length() – Counts characters.
`"Hello".length()` → 5

.toUpperCase() – Changes case.
`"java".toUpperCase()` → "JAVA"

.indexOf() – Finds text location.
`"Find me".indexOf("me")` → 5

4. Concatenation (Joining Text)

Use the + operator to stitch strings together. Be careful with spaces!

```
String fName = "John";
String lName = "Doe";
String full = fName + " " + lName; // "John Doe"
```

5. Quick Comparison

Feature	char	String
Type Category	Primitive	Non-Primitive (Object)
Quotes	Single (' ')	Double (" ")
Size	1 character exactly	0 to millions of characters
Memory	Small, fixed	Varies with text length

The Golden Rule

Use **char** for single codes or grades. Use **String** for names, sentences, or any general text. Never swap single and double quotes!